

INDEX

SL NO	TOPICS	SIGN
1.	Find the area of circle and area of triangle given three sides	
2.	Largest of three numbers	
3.	Reversing the digits of an integer	
4.	gcd (greatest common divisor) of two integers	
5.	Generating Prime numbers	
6.	Computing nth fibonacci numbers	
7.	Finding even and odd numbers	
8.	Exchanging the values of two variables	
9.	Counting Print number from 100 to 200 which are divisible by 7 and display their sum and count using for loop	
10.	Summation of set of numbers	
11.	Factorial computation	
12.	Generation of Fibonacci sequence	
13.	Array Order Reversal	
14.	Finding the maximum number in a set	
15.	Removal of an duplicates from an ordered array	
16.	Partition in an Array	
17.	Finding the kth smallest element	
18.	Read N (maximum 5) students marks and find number of student passed and failed depending on the marks	

INDEX

[illegible]

Lab Program - 01

1. Find the area of circle and area of triangle
given three sides

Program:

```
#include <stdio.h>
#include <math.h>
#define pi 3.142
int main()
{
    int r, a, b, c;
    float area_c, area_t, s;
    printf("enter radius \n");
    scanf("%d", &r);
    area_c = pi * r * r;
    printf("Area of circle = %f", area_c);
    printf("\n enter 3 sides \n");
    scanf("%d %d %d", &a, &b, &c);
    s = (a + b + c) / 2;
    area_t = sqrt(s * (s - a) * (s - b) * (s - c));
    printf("Area of triangle = %f", area_t);
    return 0;
}
```

Output

enter radius 12
Area of circle = 452.447998
enter 3 sides
14
15
12
Area of triangle = 69.282036

enter radius -1
Area of circle = 3.142000
enter 3 sides
14
15
16

Lab Program - 02

② Write a C program to find largest of three numbers

```
#include <stdio.h>
#include <conio.h>
void main()
{
    int x, y, z;
    printf("enter any three numbers \n");
    scanf("%d %d %d", &x, &y, &z);
    if (x > y && x > z)
        printf("x = %d is the largest number", x);
    scanf("
else
    if (y > x && y > z)
        printf("y = %d is the largest number", y);
    else
        printf("z = %d is the largest number", z);
    getch();
}
```

Output

Enter any three number

10 20 30

z = 30 is the largest
number

enter any three
number

40 50 60

z = 60 is the
largest number

Lab Program - 03

③ Reversing the digits of an integer

Program:

```
#include <stdio.h>
#include <conio.h>
void main()
{
    int n, rev = 0, rem;
    printf("Enter any number \n");
    scanf("%d", &n);
    while (n > 0)
    {
        rem = n % 10;
        n = n / 10;
        rev = rev * 10 + rem;
    }
    printf("The reverse of a given number is %d",
        rev);

    return 0;
}
```

Output

Enter any number 1234
Reverse of a given
number is 4321

Enter any number 5679
Reverse of a given
number is 9765

Lab Program - 04.

④ GCD of an Integer.

Program

```
#include <stdio.h>
int main() {
    int n1, n2;
    printf("Enter any two numbers\n");
    scanf("%d %d", &n1, &n2);
    while (n1 != n2)
    {
        if (n1 > n2)
        {
            n1 = n1 - n2;
        }
        else
        {
            n2 = n2 - n1;
        }
    }
    printf("gcd = %d", n1);
    return 0;
}
```

output

Enter any two number

5 10

gcd = 5

Enter any two number

1 5

gcd = 1

Lab Program - 05

Program to generate Prime Numbers

Program

```
#include <stdio.h>
#include <conio.h>
void main()
{
    int n, count = 0, i, j;
    printf("enter any number \n");
    scanf("%d", &n);
    printf("Prime numbers");
    for (i = 2; i <= n; i++)
    {
        count = 0;
        for (j = 1; j <= i; j++)
        {
if (i % j == 0)
            count++;
        }
        if (count == 2)
            printf("%d", i);
        getch();
    }
}
```

output

enter any number 5
Prime numbers 2
3
5

enter any number 11
Prime numbers 2
3 11
5
7

Program :

```
#include <stdio.h>
#include <conio.h>
int main()
{
    int n, a=0, b=1, c=0, i=1;
    printf("Enter the number\n");
    scanf("%d", &n);
    printf("The %d th fibonacci number is:", n);
    for (i=1; i<=n; i++)
    {
        a=b;
        b=c;
        c=a+b;
    }
    printf("%d\t", b);
    return 0;
}
```

Output

Enter the number 5

The 5 th fibonacci number is 3

Lab Program -07

⑦ Write a C program to find even or odd number

```
#include <stdio.h>
#include <conio.h>
void main()
{
    int num;
    printf("Enter any number \n");
    scanf("%d", &num);
    if (num % 2 == 0)
    {
        printf("%d is even", num);
    }
    else
    {
        printf("%d is odd", num);
    }
    return 0;
}
```

Output

Enter any number 2

2 is even

Enter any number 3

3 is odd

Lab Program - 08

⑧ Write a C program to exchange two values

```
#include <stdio.h>
#include <conio.h>
void main()
{
    int a, b, c;
    printf("enter any two number \n");
    scanf("%d %d", &a, &b);
    a = b;
    b = c;
    c = a;
    printf("Value after exchanging a = %d b = %d",
           a, b);
    getch();
}
```

Output

Enter any two number

a = 10 b = 20

Value after exchanging

a = 20 b = 10

Enter any two
number

a = 50 b = 10

Value after
exchanging

a = 10 b = 50

Lab Program - 09

Program :

```
#include <stdio.h>
#include <conio.h>
#include <math.h>
void main ( )
{
    int i, sum = 0, count = 0;
    printf("Numbers are :\n");
    for (i = 100; i <= 200; i++)
    {
        if (i % 7 == 0)
        {
            printf("%d\n", i);
            sum = sum + i;
            count = count + 1;
        }
    }
    printf("Sum = %d \t count = %d", sum, count);
    getch();
}
```

Output

Numbers are 105
112
119
126
133
140
147
154
161
168
175
182
189
196

Sum = 2107
count = 14

Lab Program -10

Program: C program to find the summation of set of numbers

```
#include <stdio.h>
#include <conio.h>
void main()
{
    int i, n, num, sum=0;
    printf("\n Enter the limit \n");
    scanf ("%d", &n);
    for (i=1; i<=n; i++)
        while (i<=n)
        {
            printf("Enter the number \n");
            scanf ("%d", &num);
            i=i+1; sum = sum + num;
        }
    printf(" Sum of numbers: %d", sum);
    getch();
}
```

Output

```
Enter the limit: 5
enter the number 15
enter the number 20
enter the number 30
enter the number 55
enter the number 10
Sum of numbers : 130
```


Lab Program -11

Program

```
#include <stdio.h>
#include <conio.h>
void main()
{
    int n, i, fact = 1;
    printf("Enter the value for n");
    scanf("%d", &n);
    for (i = 1; i <= n; i++)
    {
        for
        fact = fact * i;
    }
    printf("The factorial of %d", fact);
    getch();
}
```

Output

enter the value for n = 3
The factorial of 3 is : 6

Lab Program - 12.

⑫ Generation of Fibonacci Sequence.

```
#include <stdio.h>
#include <conio.h>
void main()
{
    int i, n, a=0, b=1, c;
    printf("enter the limit for fibonacci \n");
    scanf("%d", &n);
    for (i=1; i<=n; i++)
    {
        printf("%d", a);
        c=a+b;
        a=b;
        b=c;
    }
    getch();
}
```

Output

Enter the limit for
fibonacci 5

0
1
1
2
3
5

Enter the limit for
fibonacci 3

0
1
1
2
3

Lab Program - 13

```
#include <stdio.h>
#include <conio.h>
void main()
{
    int a[10], i, n;
    printf("Enter the limit:");
    scanf("%d", &n);
    printf("Enter %d numbers:", n);
    for (i = 0; i < n; i++)
        scanf("%d", &a[i]);
    printf("Reversed numbers are:");
    for (i = n - 1; i >= 0; i--)
        printf("%d\n", a[i]);
    getch();
}
```

Output

```
Enter the limit : 5
Enter 5 elements : 10
                  20
                  30
                  40
                  50

Reversed numbers are : 50
                      40
                      30
                      20
                      10
```

Program:

```
#include <stdio.h>
#include <conio.h>

void main()
{
    int a[10], n, i, j, k;
    printf("Enter the limit:");
    scanf("%d", &n);
    printf("Enter %d numbers:", n);
    for (i = 0; i < n; i++)
        scanf("%d", &a[i]);
    printf("Entered numbers are:");
    for (i = 0; i < n; i++)
    {
        for (j = i + 1; j < n; j++)
        {
if (a[i] == a[j])
if (a[i] == a[j])
        }
        for (k = j; k < n; k++)
        {
            a[k] = a[k + 1];
        }
        n--;
        j--;
    }
}
```



```

    }
    }
}

printf("After removal of duplicates:");
for (i=0; i<n; i++)
    printf("%d\n", a[i]);
getch();
}

```

Output

enter the limit : 5

enter 5 numbers : 10

11
12
13

entered numbers : 10

11
11
12
13

After removal of
duplicates

10
11
12
13

Lab Program - 16

Program

```
#include <stdio.h>
#include <conio.h>
void main()
{
    int a[10], n, i, mid = 0;
    printf("Enter the limit:");
    scanf("%d", &n);
    printf("Enter %d elements", n);
    for (i = 0; i < n; i++)
        scanf("%d", &a[i]);
    mid = (0 + (n - 1)) / 2;
    printf("Mid = %d\n", mid);
    printf("First Partition:");
    for (i = 0; i <= mid; i++)
        printf("\n %d", a[i]);
    printf("Second partition:");
    for (i = mid + 1; i < n; i++)
        printf("\n %d", a[i]);
    getch();
}
```


Output

Enter the limit : 5

enter the 5 elements: 11

22

33

44

55

entered numbers are

11

22

33

44

55

First partition is

11

22

33

Second partition is

44

55

Lab Program - 17.

⑪ Find kth smallest number in an Array

```
#include <stdio.h>
#include <conio.h>
void main() {
    int n, a[10], i, min;
    printf("enter the limit\n");
    scanf("%d", &n);
    printf("Enter %d elements:", n);
    for (i = 0; i < n; i++)
        scanf("%d", &a[i]);
    min = a[0];
    for (i = 0; i < n; i++)
    {
        if (a[i] < min)
            min = a[i];
    }
    printf("The kth smallest number in an Array is %d", min);
    getch();
}
```

Output
 enter the limit 5
 enter the 5 elements
 10 20 30 40 50
 The kth smallest number
 in an array is 10

enter the limit 3
 enter 3 elements
 50 60 70
 The kth smallest
 number in an array 50

Lab Program -18

program:

```
#include <stdio.h>
int main()
{
    int m[10], p=0, f=0, i, n;
    printf("Enter the limit \n");
    scanf("%d", &n);
    printf("enter marks \n");
    for (i=0; i<n; i++)
    {
        scanf("%d", &m[i]);
    }
    for (i=0; i<n; i++)
    {
        if (m[i] >= 40)
            p = p + 1;
        else
            f = f + 1;
    }
    printf("pass = %d & fail = %d", p, f);
    return 0;
}
```

Output: enter the limit 5	
enter marks	10
	20
	50
	60
	80
pass = 3	fail = 2

Lab Program -19

Program :

```
#include <stdio.h>
#include <conio.h>
#include <string.h>
void main()
{
    char s[40];
    int i, consonants=0, vowels=0, special=0, digits=0;
    printf("Enter the string:");
    gets(s);
    for (i=0; i=strlen(s); i++)
    {
        if ((s[i]>='a' && s[i]<='z') || s[i]>='A' && s[i]<='Z'))
        {
            if (s[i]== 'a' || s[i]== 'e' || s[i]== 'i' || s[i]== 'o' ||
                s[i]== 'A' || s[i]== 'E' || s[i]== 'I' || s[i]== 'O' ||
                s[i]== 'u')
            {
                vowels++;
            }
            else
            {
                consonants++;
            }
        }
        else if (s[i]>='0' && s[i]<='9')
        {
            digits++;
        }
        else
        {
            special++;
        }
    }
}
```



```
{  
    special++;  
}  
}  
  
printf("Vowels: %d \n", vowels);  
printf("Consonants: %d \n", consonants);  
printf("Digits: %d \n", digits);  
printf("Special characters: %d", special);  
getch();  
}
```

Output

Enter the string : SCS college, Bhatkal 8971071471
Vowels : 5
Consonants : 12
Digits : 10
Special characters : 2

Lab Program - 20

Program

```
#include <stdio.h>
#include <conio.h>
void main()
{
    int a[3][3], b[3][3], s[3][3], d[3][3], r, c;
    printf("Enter the Order of Matrix\n");
    scanf("%d %d", &r, &c);
    printf("Enter Matrix A elements\n");
    for (i = 0; i < r; i++)
    {
        for (j = 0; j < c; j++)
        {
            scanf("%d", &a[i][j]);
        }
    }

    printf("Enter matrix B elements\n");
    for (i = 0; i < r; i++)
    {
        for (j = 0; j < c; j++)
        {
            scanf("%d", &b[i][j]);
        }
    }

    for (i = 0; i < r; i++)
    {
        for (j = 0; j < c; j++)
        {
            s[i][j] = a[i][j] + b[i][j];
            d[i][j] = a[i][j] - b[i][j];
        }
    }
}
```



```
printf("Sum of Matrices is \n\n");  
for(i=0; i<r; i++)
```

```
{  
    for(j=0; j<c; j++)
```

```
{  
    printf("%d\t", s[i][j]);
```

```
;  
    printf("\n");
```

```
;  
printf("Subtraction of Matrices is \n\n");
```

```
for(i=0; i<r; i++)
```

```
{  
    for(j=0; j<c; j++)
```

```
{  
    printf("%d\t", d[i][j]);
```

```
;  
    printf("\n");
```

```
;  
    getch();
```

```
;  
}
```

Output

Enter the order of matrix 2 2

Enter Matrix A elements 6 3 4 2

Enter Matrix B elements 2 1 3 1

Sum of matrices 8 4

7 3

Subtraction of Matrices is 4 2

1 1

Seen
16/12/20